

Two Array Average

Time limit: 1.00 s

Memory limit: 512 MB

You are given two arrays of n integers.

Your task is to select a nonempty prefix from both arrays such that the average of all selected numbers is as large as possible.

Input

The first line has an integer n .

The second line has n integers $a_1, a_2, \dots, a_n$: the numbers in the first array.

The third line has n integers $b_1, b_2, \dots, b_n$: the numbers in the second array.

Output

Print two numbers: the prefix sizes.

Your answer is considered correct if the absolute or relative difference to the maximum average is at most 10^{-6} .

Constraints

- $1 \leq n \leq 10^5$
- $1 \leq a_i, b_i \leq 10^9$

Example

Input	Output
4 1 5 5 2 3 1 3 1	3 1

Explanation: if you choose the prefixes $[1, 5, 5]$ and $[3]$, the average is $(1 + 5 + 5 + 3)/4 = 3.5$ which is the maximum possible average.